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REMOVING INFANT RHESUS FROM MOTHER FOR 13 DAYS COMPARED WITH REMOVING MOTHER FROM INFANT

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INTRODUCTION

A NUMBER of studies have shown that temporary separation from the mother may affect the behaviour of infant rhesus monkeys for some time after reunion with her (refs. in Hinde and Spencer–Booth, 1971). In most of these experiments the separation involved temporary removal of the mother, the infant remaining in the same physical and social environment in which it had lived previously: in this way, any additional effects which might arise if the infant were moved temporarily to a strange environment were avoided. In the present experiment the procedure was reversed, the infant being removed to a strange cage and the mother left in her familiar environment. It was predicted that the effects on the infant of this treatment, involving both separation from the mother and the trauma of a strange environment, would be more severe than those of mere removal of the mother.

MATERIALS AND METHODS

Subjects

The experiments were performed with rhesus monkeys living in small captive groups. Each group consisted of an adult male, 2–4 females and their young. Five mother–infant pairs were used, the infants (3 female and 2 male) being 30-to 32-weeks-old at the time they were removed from their mothers. Data on these "infant-removed" infants are compared with those from 6 (4 female and 2 male) "mother-removed" infants as published previously (Spencer-Booth and Hinde, 1971).

Accommodation

Each group occupied an outside pen (5·50 × 2·44 × 2·44 m) connecting with an indoor room (2·29 × 1·83 × 1·37 m) (see also Hinde and Rowell, 1962). While separated from their mothers, the infants were kept in a cage (1·13 × 1·00 × 1·18 m) in an indoor room about 50 m from and out of sight of the group: they were visually but not auditorally isolated from other monkeys in the room. The latter were caged singly or in pairs, and were not undergoing separation at the time.

Methods

Each infant was removed from its mother when between 30- and 32-weeks-old and placed in the indoor cage for 13 days, after which it was returned to its mother

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in the group cage. The data used in this analysis were concerned only with 3–4 days before the infant's removal, the separation period, and 4 weeks after the infant's return.

At 10·00 hrs on 3 of the 4 days before the separation, routine data on mother–infant interactions and the infant's activity while off its mother were recorded (see Spencer-Booth and Hinde, 1967). These watches were continued until the infant had been off its mother for 2 hr or until 14·30 hrs, whichever was earlier.

At 10·00 hrs on day 4, the infant was caught in the outside pen and taken to the indoor room, where data on activity and vocalizations were recorded for 2 hr. These watches were repeated on days 5, 9, 12, 15 and 16. At 10.00 hrs on day 17, the infant was returned and data collection began immediately as on days 1–3. Similar data were collected on days 18, 19, 21, 23, 30, 37 and 44.

For observations in the home pens the observer was in a small hut, 7·7 m from the end of the pen, and was visible to the monkeys through a window. The monkeys were thoroughly habituated to this procedure, and there was no reason to believe that the observer's presence influenced their behaviour in any way. For observations in the indoor cage the observer was seated about 2 m from and in full view of the infants: in this case the animal's behaviour could be affected by sudden movements by the observer.

More limited data on the mothers were also collected. On one of the four pre-separation days, and on days 4 and 15 within the separation period, the mother's locomotor activity and vocalizations were recorded from 10·00 to 12·00 hrs.

Measures

The following were used in this analysis:

Time off. The percentage of half min watched in which the infant was off the mother and all other animals.

The relative frequency of rejections ($R/(A + M + R)$). Every time the infant got on or tried to get on the mother's nipple we recorded whether it was through the mother's initiative (M) or the infant's: in the latter case the infant could be accepted (A) or rejected (R). From this was calculated the proportion of the total number of times the infant got on the nipple, or attempted to do so, which were rejected.

Infant's role in making successful nipple contacts. This is given by the ratio $A/(A + M)$.

Infant's role in breaking nipple contacts. When the infant came off the nipple we recorded whether it was the infant's initiative (I'), the mother's initiative (M') or mutual. The ratio $I'/(I' + M')$ gives a measure of the infant's role.

Time at a distance from mother (> 60 cm). The percentage of half min in which the infant was off in which it was recorded more than 60 cm from the mother (60 cm is roughly the distance a mother can reach in a hurry, and was assessed in relation to the 120 cm divisions of the wire netting).

Infant's role in maintaining proximity ($\%Ap - \%L$). The percentage of occasions that the distance between mother and infant decreased from more than 60 cm to less than 60 cm (approaches) or increased from less to more (leavings), that were due to movement by the infant, were calculated. The difference between the proportions of approaches and of leavings which were due to movement by the infant gives a measure of the infant's role in maintaining proximity which is relatively

independent of the absolute levels of activity of the two animals (Hinde and Atkinson, 1970).

Whoo calls. The mean number of whoo calls per hundred half min in which the infant was off its mother (infant) or per hundred half min (mother).

Locomotor activity. The outside pen was subdivided into 16 "boxes" (upper and lower sections and eight equal floor areas) and the indoor cage into six. Locomotor activity was assessed in terms of the number of boxes entered per half min watched for the mother, and per half min off the mother for the infant.

Activity when active ($> 5/> 1\%$). The number of half min in which the infant or mother entered > 5 boxes, as a percentage of the number in which it entered more than one.

Sitting ($\%S$). The number of half min in which the animal was recorded sitting still and not engaged in any obvious activity, as a percentage of the number in which it was off its mother (infant) or the number watched (mother).

Rough-and-tumble social play ($\%R\&T$). The number of half min in which the infant engaged in rough-and-tumble social play, as a percentage of the number it was off its mother.

Approach/withdrawal social play ($\%A/W$). The number of half min in which the infant engaged in approach/withdrawal social play, as a percentage of the number it was off its mother.

Method of analysis

Data were analysed separately for each animal for each watch, and were grouped as follows. The mean of the data was determined for days 1–3, days 18–23 and days 30–44, for each infant. The effects of the separation period were assessed by analysing the data for each day of that period (days 4, 5, 9, 12, 15, 16) separately. For changes between periods, Sign tests (two-tailed) were used, while for comparisons between groups, Mann–Whitney U tests (two-tailed) were employed.

It must be emphasized that comparisons of infants or of mothers between infant-removed and mother-removed groups during the separation period are limited by the difference in size between the outside runs and the indoor cages.

RESULTS

Comparison of groups before separation

The mother-removed and infant-removed groups did not differ significantly on any of the measures of mother–infant interaction, or of infant behaviour when off mother.

Behaviour of the infants during separation (Fig. 1)

Initial hypoactivity was shown by the two males (Whisky and Matthew). On release into the indoor cage they immediately climbed into a top corner and stayed there for most of the 2 hr watch. They first moved only after 1 hr 21 min and 1 hr 45 min respectively, and when they did so their movements were slow and cautious. Matthew did not call at all, while Whisky only gave soft whoo calls when other animals in the room called. By the next day, Matthew had just started to call, while Whisky's calling had decreased from that of the previous day. They were both moving

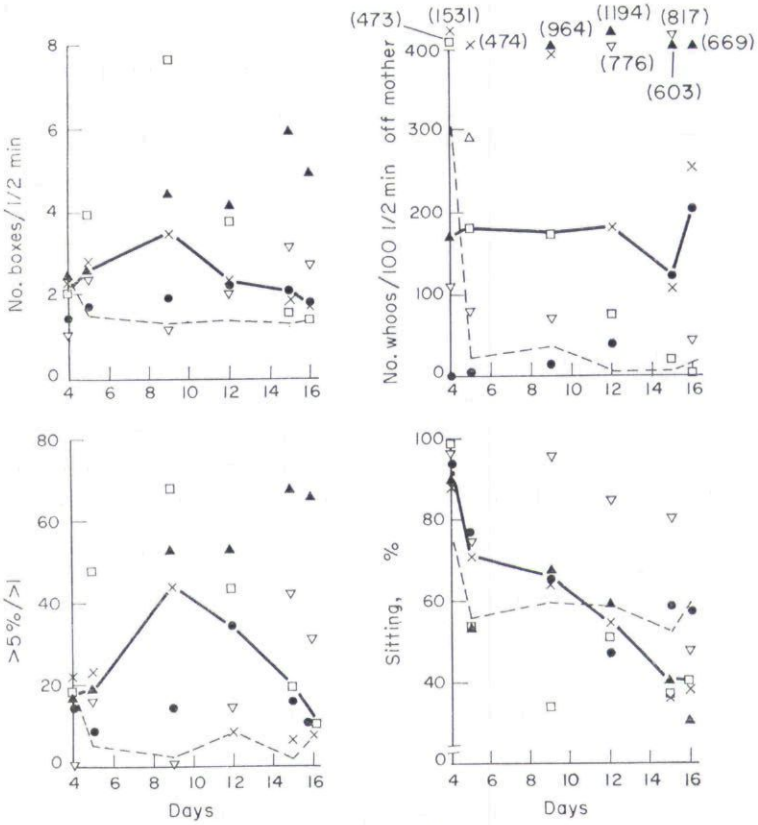


FIG. 1. Behaviour of infants during the separation period.

The symbols indicate individual infant-removed infants, the closed circle and the inverted triangle representing the data for the two males. The continuous line shows the medians for the infant-removed infants, the discontinuous one those for mother-removed infants obtained in the earlier study. Top left—locomotor activity, measured by the number of "boxes" entered per half min. Top right—whoo calling. Bottom left—activity when active. Bottom right—percentage of half min sitting doing nothing .

around more. On day 9, Whisky seemed to be suffering from exhaustion: his activity and calling had decreased, and although he was still whooing fairly often, he was eating very little, moving around slowly and sometimes falling over. Matthew's activity and calling continued to increase steadily. By day 12, Whisky had recovered considerably; he was more active and his calling had increased markedly, but he still spent much time inactive. Matthew showed a further increase in activity and calling. By day 15, the calling of both had increased again, but Matthew's activity had decreased very slightly. On day 16, Whisky's calling had decreased considerably but he was still quite active. He clung to the cage bars and flung his head back. Matthew's calling had increased, but his activity had decreased slightly and he was nail-biting and thumb-sucking.

Although one of the females was immobile for 20 min after release into the indoor cage, the others became active immediately. All three thereafter whooed frequently,

and were active for much of the time, though they also frequently sat inactive. They sometimes growled at the observer. One made frantic scrabbling movements along the sides of the cage and another frequently ground her teeth. On days 5 and 9 all three showed a progressive increase in locomotor activity, accompanied in two cases by some decrease in calling. By day 12, activity was decreasing and they spent longer periods eating. The activity of two of these females decreased further by days 15 and 16, and they showed signs of depression. One sat huddled to the wall for much of the time. All three were noticeably unresponsive to the observer and to room noises. One, however, still scrabbled intermittently at the sides of the cage, and thus had a high activity score, and still called often. All three showed some stereotyped pacing of the cage. By this time all five infants had the typical depressed physical appearance of separated infants. Their coats were raised and matted, their cheeks sunken and bodies hunched.

The individual differences in the pattern of responding over the separation period, as well as the differences in cage-size, limit the usefulness of comparisons with infants of the mother-removed group during the separation period. In general the sorts of behaviour shown were qualitatively similar in the two groups, but the fear and associated immobility of some of the infant-removed infants was more marked than anything seen in the mother-removed infants. Furthermore, the searching behaviour and/or hyperactivity was more marked and longer-lasting in the infant-removed group: perhaps for this reason the onset of depression was delayed.

To be more specific, the initial median frequency of whoo calling was somewhat higher in the mother-removed group, owing largely to the hypoactivity of the two males in the infant-removed group. However the difference was not significant: indeed, one of the infant-removed infants called over twice as often as any other infant in either group on day 4. Subsequently the situation was reversed, the medians for the mother-removed group being lower (day 12, $p = 0{\cdot}03$; day 15, $p = 0{\cdot}004$; days 5, 9 and 16, *n.s.*).

Comparisons of locomotor activity between the two groups are especially liable to be influenced by the differences in cage size, but it is noteworthy that, while the median number of boxes entered was (not-significantly) higher for the mother-removed infants on the first day of separation, it was lower on all subsequent days (days 5 and 12, $p = 0{\cdot}004$; day 15, $p = 0{\cdot}03$; days 4, 9 and 16, *n.s.*; Mann Whitney U test, two-tailed).

The proportion of minutes in which the infants were recorded sitting showed, as might be expected, opposite trends. At first it was higher for the infant-removed group (day 4, $p = 0{\cdot}02$; days 5 and 9, *n.s.*), but subsequently non-significantly lower. The decrease in %S with time was not due merely to the increase in locomotor activity of the two males, but was characteristic of all individuals (Spearman rank order correlation with serial order of observation negative in all cases, $p < 0{\cdot}05$ in two).

In conclusion, although individual differences amongst the infant-removed infants were marked, their behaviour during separation involved the same sort of changes as were found in the mother-removed group, but with considerable differences in degree. In particular, two animals showed initial fear and immobility, all later showed hyperactivity, and symptoms of depression appeared rather later.

Behaviour of mother during separation (Fig. 2)

Data were collected from four mothers on day 4 and from all five before separation and on day 14. While none gave whoo-calls during the pre-separation watch, all four watched on day 4 did so. They seemed extremely restless, alternating periods of sitting (%S was higher than pre-separation in all four) with periods of activity (No. of boxes entered higher than before separation in 3 out of 4 animals, and activity when active in all 4).

On day 14 whoo-calling had decreased virtually to zero. The frequency with which they were recorded sitting tended to be lower than on day 4 (3 out of 4 animals), but still higher than before separation (4 out of 5). Similarly, locomotor activity tended to be less than on day 4 (3 out of 4 animals) but still higher than before separation (4 out of 5).

Comparison with mothers from the mother-removed group are possible only on day 4. The mother-removed mothers were recorded sitting more than the infant-removed mothers ($p = 0{\cdot}03$), tended to be less active (*n.s.*), and were less active when active ($p = 0{\cdot}01$). However the frequency of calling showed no clear difference between the groups. In general, in so far as the differences in cage size permit comparison, the mother-removed mothers appeared to be more disturbed during separation than were the infant-removed mothers.

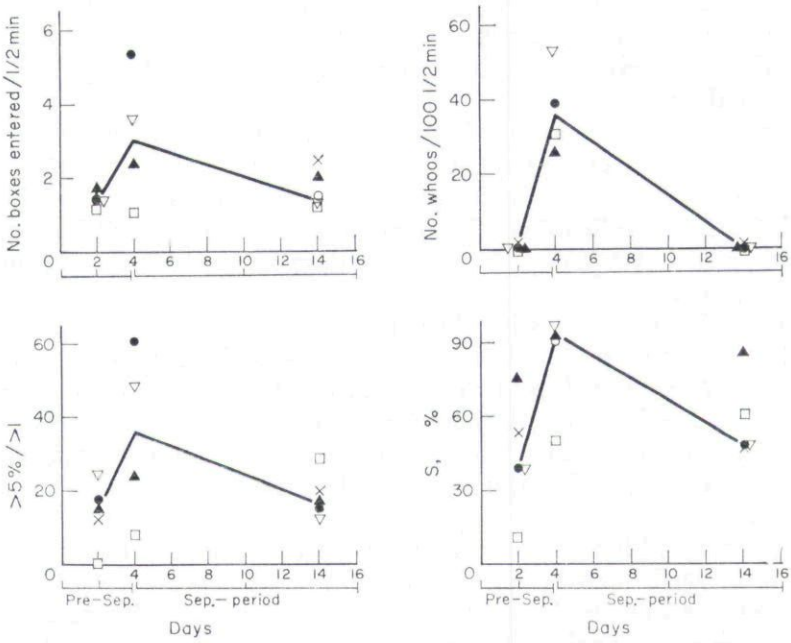


FIG. 2. Behaviour of mothers of infant-removed infants before and during separation. Symbols used for mothers are those used for their own infants in Fig. 1. Measures as in Fig. 1.

Infant behaviour after reunion (Fig. 3)

After reunion the behaviour of the infant-removed infants differed from their pre-separation behaviour in ways similar to, but less marked than, that seen in mother-removed infants. In no period did whoo-calling differ significantly from the pre-separation level. Locomotor activity was at first slightly lower, the difference for both the mean number of boxes entered per half min and for activity when active

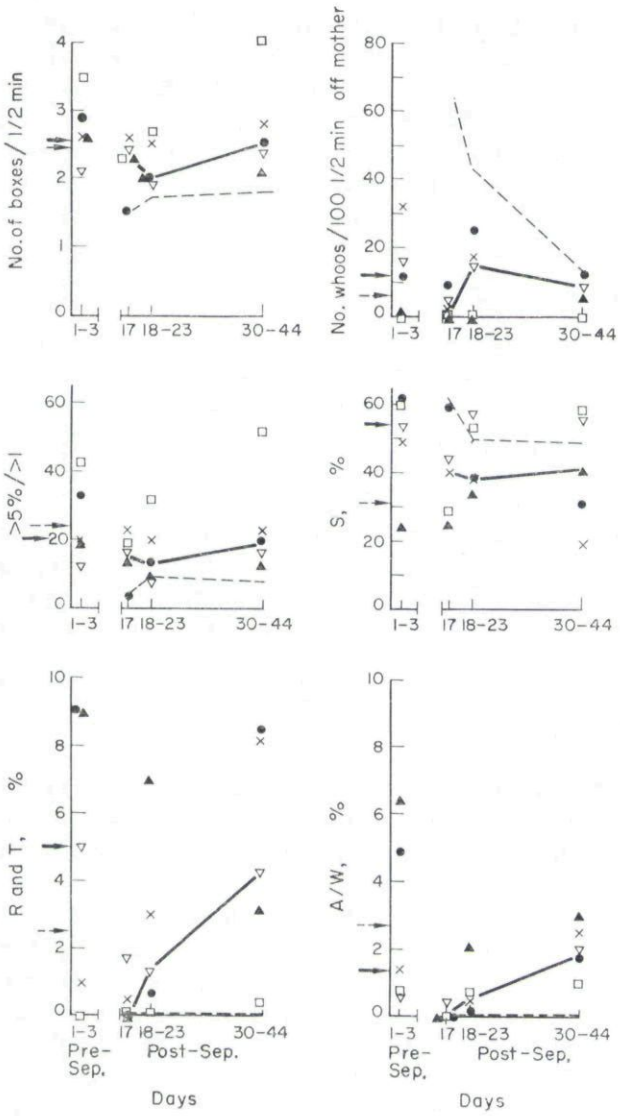


FIG. 3. Behaviour of infants before and after the separation period.

Symbols as in Fig. 1. Measures in top 4 graphs as in Fig. 1. Bottom left—percentage of half min in which rough-and-tumble play occurred. Bottom right—percentage of half min in which approach/withdrawal play occurred. The thick arrow indicates the preseparation Median for the infant-removed infants and the thin arrow that for the mother-removed infants.

being significant for days 18–23 ($p = 0{\cdot}06$, sign test): this difference had disappeared by days 30–44. The percentage of min in which they were recorded sitting tended to be similar to that before separation in all periods. Although social play was somewhat depressed in the early post re-union periods in most individuals, again in no period (except for approach/withdrawal on day 17) was the difference from pre-separation level significant.

The mildness of the effects of separation on the infant-removed infants represents a marked contrast to the mother-removed infants. The former whoo-called less than the mother-removed infants in all three post re-union periods ($p = 0{\cdot}004$, $p = 0{\cdot}03$ and *n.s.* resp.). They also showed more locomotor activity in all three periods (No. boxes, $p = 0{\cdot}04$, $0{\cdot}03$ and $0{\cdot}01$; activity when active, *n.s.*, *n.s.* and $p = 0{\cdot}004$). There were no significant differences in the percentage of minutes in which they were recorded sitting, but the infant-removed infants played significantly more than the mother-removed ones (approach-withdrawal play, *n.s.*, $p = 0{\cdot}004$ and $0{\cdot}004$; rough and tumble play, *n.s.*, $p = 0{\cdot}008$ and $0{\cdot}02$).

Mother–infant interaction after reunion (Fig. 4)

After reunion the mother–infant relationship at first differed from that obtaining before separation in ways similar to, but less marked than, that seen in mother-removed infants. The proportion of time off the mother was lower on the day of reunion than before separation in all five cases ($p = 0{\cdot}06$, Sign test, two-tailed), but the median was near or above pre-separation levels in the two later post re-union periods. The median proportion of time off which the infant spent more than 60 cm from the mother remained near the pre-separation level throughout, though it tended to be low in two individuals on days 18–23. The mothers rejected few of the infants' attempts to gain the nipple, the relative frequency of rejections being below pre-separation levels ($p = 0{\cdot}06$, $0{\cdot}06$ and *n.s.*), but the relative responsibility of infant and mother for nipple contact was little changed. On the day of reunion the infants played a smaller role in maintaining proximity to the mothers ($p = 0{\cdot}06$), though subsequently the median value of this measure returned to close to its pre-separation level.

In a number of respects the post-reunion mother–infant relationships of the infant-removed infants differed from those of the mother-removed ones. The infant-removed infants tended to spend more time off their mothers (*n.s.*, *n.s.* and $p = 0{\cdot}052$) and more of that time at a distance from them (*n.s.*, *n.s.* and $p = 0{\cdot}008$). These differences could have been due to differences between the mothers of the two groups, to differences between the infants, or both (Hinde, 1969). Some measures relevant to the relative roles of mothers and infants in determining time off the mother are given in Table 1. The initial reduction in time off the mother after reunion in the infant-removed infants was associated with a decrease in the relative frequency of rejections, but no significant change in the infant's role in making or breaking nipple contacts. It was thus due primarily to greater receptiveness by the mothers to the infants' demands for nipple contacts. By contrast, in the mother-removed group the frequency of rejections and the infants' role in making nipple contacts changed little, but there was a marked decrease in the infants' role in breaking nipple contacts. Here, therefore, the decrease in time off was due primarily

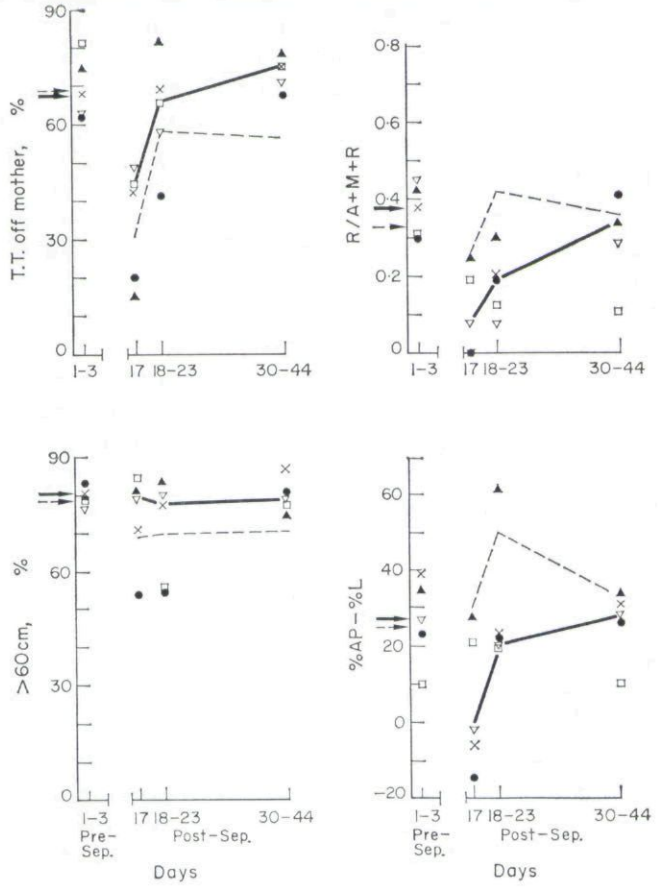


FIG. 4. Mother–infant interaction after reunion.

Symbols as in Figs. 1 and 3. Top left—total time off mother. Top right—relative frequency of rejections. Bottom left—proportion of time off mother spent at a distance from her. Bottom right—infants' role in maintaining proximity to mother when off her.

TABLE 1. RELATIVE ROLES OF MOTHER AND INFANTS IN NIPPLE CONTACTS

		Period (days)			
		Pre-separation	17	18–23	30–44
$R/(A + M + R)$	Infant-removed	0·38	0·08*	0·30*	0·34
	Mother-removed	0·33	0·26	0·42	0·31
	Sig. of inter-group difference, p	*n.s.*	$<0·06$	$<0·01$	*n.s.*
$A/(A + M)$	Infant-removed	0·94	0·90	0·92	1·0
	Mother-removed	0·89	0·92	0·93	0·92
	Sig. of inter-group difference, p	*n.s.*	*n.s.*	*n.s.*	$<0·01$
$I'/(I' + M')$	Infant removed	0·58	0·95	0·47	0·55
	Mother-removed	0·52	0**	0·06**	0·20**
	Sig. of inter-group difference, p	*n.s.*	$<0·09$	$<0·01$	$<0·02$

Median values for Infant-removed and Mother-removed groups. * and ** indicates difference from pre-separation value with $p < 0·06$ and $p < 0·02$ respectively (Sign test).

to a decrease in the role of the infants in breaking contact once they were on the nipple: the mothers did not become more receptive in the face of the infants' demands and played a greater role in terminating nipple contacts.

Similar arguments can be applied to the time spent at a distance from the mother. The non-significant change in the infant-removed group was associated with a decrease in the infants' role (i.e. increase in the mothers' role) in maintaining proximity. The slight decrease in the mother-removed group was associated with a (non-significant) tendency for the infants' role in maintaining proximity to increase. However, the differences between the two groups in the infants' role in maintaining proximity did not reach significance.

In summary, the post-reunion behaviour of the infant-removed group was characterized by fewer maternal rejections and a greater maternal role in maintaining proximity, while that of the mother-removed infants was characterized by a smaller maternal role in nipple contact and (non-significantly) in maintaining proximity. The lesser degree of disturbance in the infant-removed group was thus associated with a more positive role by their mothers in mother–infant interaction after reunion.

DISCUSSION

Infants removed from both their mothers and the environment with which they were familiar showed changes in behaviour which were rather different from those shown by infants left in their familiar environment when their mothers were removed. While the latter showed a reduction in locomotor activity, some of the former tended to show first hyperactivity, and in association with this the onset of depression was delayed. Although differences in environmental conditions during separation make direct comparisons between the two groups difficult, it is probable that, in the absence of reunion, the infant-removed infants would have been the more profoundly disturbed.

However after reunion the reverse was the case. While the mother-removed infants continued to show obvious signs of disturbance for at least several weeks, the behaviour of the infant-removed infants soon returned to levels similar to those seen before separation. At least three types of explanation could be suggested for this difference:

(i) The effects of the experience on the infants are different in *kind* in the two groups, and produce more enduring effects in the mother-removed than in the infant-removed infants. While this possibility cannot be ruled out, it seems improbable in that both groups experienced separation from the mother and the infant-removed infants experienced the strange pen as well. A difference in the opposite direction would therefore be predicted, except as indicated in (ii) below.

(ii) Apart from the trauma associated with catching, the infant-removed infants had no unpleasant experience in the home pen. while the mother-removed infants lived there throughout the separation period. The latter could therefore have become adversely conditioned to the home environment, while the former did not and therefore behaved normally as soon as they were returned to it. This possibility cannot be ruled out by our data: if it does represent an important issue, it indicates at least

that the trauma of separation can be largely overcome by suitable subsequent experience.

(iii) Post re-union behaviour could depend primarily on the degree of disturbance suffered by the mothers. This was almost certainly much less in the infant-removed group. In harmony with this view are the findings that, (a) the smaller degree of disturbance shown by the infant-removed infants after reunion was associated with a more rapid restoration of mother–infant relations, and (b) this involved a greater maternal role in mother–infant interaction.

Thus these findings strongly suggest that the effects sometimes produced on infant behaviour and development by a period of separation from the mother may be due not (or not only) to separation *per se*, as has often been suggested for both human and subhuman primates, but (also) to the associated disturbance of the relationship between the infant and the individual who has hitherto formed the most important part of its environment.

SUMMARY

In a previous experiment, 6 rhesus monkey mothers were removed from their infants for 13 days, the infants being left in their familiar environment. Here the procedure was reversed—5 infants were removed to an isolation cage in a strange room, the mothers staying in their familiar environment. Contrary to expectation these infant-removed infants were less disturbed than the mother-removed infants. This difference was associated with, and was probably due to, a more positive role by the mothers of the infant-removed group in mother–infant interaction after reunion. It is suggested that the effects of a separation experience may be mediated in part by the disturbance to the mother–infant relationship which results.

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